

Hichem Rami AIT EL HARA

PhD in Computer Science · R&D Engineer · Formal Methods & SMT Solving

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SUMMARY

Formal methods researcher and engineer, with 5 years of experience working at a French R&D consultancy, conducting studies, implementing prototypes and delivering solutions to industrial and academic clients and partners. Author and co-author of multiple publications at venues including PLDI, TACAS, and LPAR, and contributor to multiple open-source projects.

SKILLS

Formal Methods: SMT Solving, Constraint Programming, Deductive Verification, Symbolic Execution, Code Transformation, Abstract Interpretation

Software Engineering: Algorithm and Data Structure Design and Implementation, Functional Programming, Fuzz Testing

Programming Languages: OCaml (Expert); Python, Lean, Rust (Novice); C, Rocq, Java, Scala (From school)

Development Tools: Git, VS Code, Unix, LaTeX, Typst (Experienced); Nix (Novice)

Languages: English (Fluent), French (Fluent), Kabyle (Native), Arabic (Fluent), Spanish (Basic)

EXPERIENCE

- OCamlPro** [↗](#), R&D Engineer Sept 2021 – present
- Conducted studies, implemented prototypes and delivered solutions for industrial and academic clients and partners
 - Contributed to multiple open-source projects ([Alt-Ergo](#) [↗](#), [Owi](#) [↗](#), [Colibri2](#) [↗](#), [Smtml](#) [↗](#), [Dolmen](#) [↗](#), ...)
 - Managed relationships with clients and prospects, organized and led technical meetings, and secured funding for R&D projects
- Université Paris-Saclay** [↗](#), **CEA List** [↗](#), PhD Student (CIFRE PhD, jointly with [OCamlPro](#) [↗](#)) Sept 2022 – Oct 2025
- Designed an SMT theory of sequences tailored for the verification of Ada programs
 - Authored and co-authored multiple research papers (see Publications)
 - Implemented multiple calculi to reason over SMT problems using (n-indexed) sequences and a difference logic engine
 - Developed the labeled union-find data structure and used it both to reason over SMT sequences and to implement a new interval domain for real and integer arithmetic
- OCamlPro** [↗](#), R&D Intern Mar 2021 – Sept 2021
- Developed [Alt-Ergo-Fuzz](#) [↗](#), a fuzzer for the [Alt-Ergo](#) [↗](#) SMT solver, published at JFLA 2022
 - Implemented context reinitialization in [Alt-Ergo](#) [↗](#), making it possible to reuse one running instance for multiple problems
 - Found both runtime and subtle soundness bugs, most of which have been fixed since

EDUCATION

- PhD** **Université Paris-Saclay** [↗](#), **CEA List** [↗](#), Computer Science Sept 2022 – Oct 2025
- Thesis: [Theory of Sequences Tailored for Program Verification](#) [↗](#) (CIFRE PhD, jointly with [OCamlPro](#) [↗](#))
 - [Defended on 15 October 2025](#) [↗](#)

MSc [Sorbonne University](#), Computer Science

Graduated Sept 2021

- [Software Science and Technology \(STL — Science et Technologie du Logiciel\)](#), R&D track focused on Software safety and reliability
- Main subjects: Algorithmics, Formal Methods, Type Checking, Static Analysis, Program Verification, Operational Semantics and Process Algebras

BSc [Sorbonne University](#), Computer Science

Graduated Sept 2019

PUBLICATIONS

colibri2: A CP solver for SMT problems

June 2026

François Bobot, **Hichem Rami Ait-El-Hara**, Bruno Marre
[cea.hal.science/cea-05644448](#) (Preprint)

Smt.ml: A Multi-Backend Frontend for SMT Solvers in OCaml

Apr 2026

João Madeira Pereira, Filipe Marques, Pedro Adão, **Hichem Rami Ait-El-Hara**, Léo Andrès, Arthur Carcano, Pierre Chambart, Petar Maksimović, Nuno Santos, José Fragoso Santos
[link.springer.com/chapter/10.1007/978-3-032-22752-2_2](#) (TACAS @ ETAPS 2026)

Theory of Sequences Tailored for Program Verification

Nov 2025

Hichem Rami Ait-El-Hara

[theses.fr/2025UPASG067](#) (PhD Dissertation)

Reasoning over n-Indexed Sequences in SMT

Aug 2025

Hichem Rami Ait-El-Hara, François Bobot, Guillaume Bury

[link.springer.com/article/10.1007/s00236-025-00496-w](#) (Acta Informatica — Selected Extended Papers of SMT 2024 and Related Papers)

Constraint Propagation for Bit-Vectors in Alt-Ergo

Aug 2025

Hichem Rami Ait-El-Hara, Guillaume Bury, Basile Clément, Pierre Villemot

[ceur-ws.org/Vol-4008](#) (SMT 2025: 23rd International Workshop on Satisfiability Modulo Theories)

Relational Abstractions Based on Labeled Union-Find

June 2025

Dorian Lesbre, Matthieu Lemerre, **Hichem Rami Ait-El-Hara**, François Bobot

[dl.acm.org/doi/10.1145/3729298](#) (PLDI 2025: 46th ACM SIGPLAN Conference on Programming Language Design and Implementation)

An SMT Theory for n-Indexed Sequences

July 2024

Extended abstract; see the full paper above (Reasoning over n-Indexed Sequences in SMT, Acta Informatica)

Hichem Rami Ait-El-Hara, François Bobot, Guillaume Bury

[ceur-ws.org/Vol-3725](#) (SMT 2024: 22nd International Workshop on Satisfiability Modulo Theories)

On SMT Theory Design: The Case of Sequences

May 2024

Hichem Rami Ait-El-Hara, François Bobot, Guillaume Bury

[easychair.org/publications/paper/qdvj](#) (LPAR 2024: 25th Conference On Logic For Programming, Artificial Intelligence and Reasoning)

Alt-Ergo-Fuzz: A fuzzer for the Alt-Ergo SMT solver

June 2022

Hichem Rami Ait-El-Hara, Guillaume Bury, Steven de Oliveira

[hal.inria.fr/hal-03626861](#) (JFLA 2022: 33èmes Journées Francophones des Langues Applicatifs)

TALKS & SERVICE

See [Talks](#) and [Service](#) for a full list of invited talks, presentations, and reviewing/committee activities.